

Bridging Digital Wellbeing and Accessibility: An Analysis of the W3C Web Content Accessibility Guidelines

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Abstract

The W3C Web Content Accessibility Guidelines (WCAG), first released in 1999, have over time expanded their focus on cognitive and learning disabilities, including attentional needs. User attentional needs are also a focus of digital wellbeing, a research area which aims to help users spend their time well when using technology, reducing overuse and increasing meaning. However, the overlap and potential synergies between WCAG and DWB have not yet been measured. We use an expert coding method to classify all requirements from the main versions of WCAG (1.0, 2.2, and 3.0 draft) with respect to their secondary effects on digital wellbeing and attention-capture dark patterns. We note that an increasing proportion of WCAG requirements promote digital wellbeing, although in some cases WCAG may play a merely enabling role, a limited role, or a negative role. Our findings bring forward a link between accessibility and digital wellbeing which has not been previously established.

CCS Concepts

• **Human-centered computing** → **Accessibility design and evaluation methods; HCI design and evaluation methods; Interaction paradigms.**

Keywords

Digital wellbeing, WCAG, accessibility, digital self-control tools, digital overuse, persuasive technology

ACM Reference Format:

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1 Introduction

The W3C Web Content Accessibility Guidelines (WCAG), first released in 1999, aim to improve the accessibility of web content

for people with disabilities. This specification, targeting “visual, auditory, physical, speech, cognitive, language, learning, and neurological disabilities” [1] provides various requirements which can be used to evaluate the accessibility compliance of a webpage. Given the wide global implementation of WCAG in government and non-government web interfaces [2], WCAG serves as a particularly comprehensive method of considering best practices for how accessible web pages around the world should be designed. Over time, the number of total requirements within the specification has increased from 65 in 1999 (version 1.0) to 86 in 2023 (version 2.2) to 180 planned requirements (draft version 3.0) [1, 3, 4], with a consistently increased focus on cognitive and learning disabilities. One particular target of accessibility for cognitive disabilities, as mentioned by the W3C, is the capacity of the webpage to support the user’s attention [5].

Users’ attention is also a focus area of digital wellbeing (DWB) studies in human-computer interaction. Digital wellbeing has been conceptualized by Burr et. al in 2020 as “the impact of digital technologies on what it means to live a life that is good for a human being in an information society” [6]. The existing applied DWB research theme has mainly focused on the overuse of technology in the leisure space (i.e. discretionary and undesired use of applications) [7, 8]. Within digital wellbeing, tools called DSCTs (digital self-control tools) have been created to help users control the quality or quantity of their time spent when using leisure applications such as social media, entertainment, or video sites [8]. The underlying driver for the digital wellbeing approach in the DSCT context is the (internal) motivation of users to spur behavior change when confronted with their use of digital technology [8, 9]. The change motivation is satisfied by the use of a DSCT which instigates a change in a user’s digital environment (block/removal, self-tracking, goal-advancement, or reward/punish) [7]. Such digital interventions, suitable for certain web, desktop, and mobile devices, are intended to lead to secondary effects on the user’s behavior [8, 9].

The digital wellbeing and digital self-control tool areas represent a unique method of empowering leisure app users through technology, ultimately supporting user autonomy based on the question “How do you want to spend your time?” This method of transforming reflective user needs into practical technology builds on previous theoretical work in HCI such as meaningful HCI and positive computing [10, 11]. In 2023 a systematic digital wellbeing literature review identified specific aspects, themes, intervention types, behavioral theories, and challenges in the digital wellbeing area, rigorously defining the current research context while also

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analyzing digital self-control tools [8]. In particular, *attention*, *attention capture design patterns*, the *attention economy*, and *engagement* continue to be key focus areas for the DWB study area [8].

Acknowledging that the WCAG specification has been known to cause secondary effects on usability, as noted by WCAG and other authors [12–14], an expert coding method was used to classify 331 total design requirements from the WCAG 1.0, 2.2, and 3.0 (draft) standards to understand if these design requirements also have a potential secondary effect on the digital wellbeing of users (as defined in the 2023 systematic DWB literature review) [8]. Also coded was the distribution of which attention-capture dark patterns (as defined in [15]) are affected under the *Avoiding attention-capture dark patterns* DWB aspect for these requirements.

Our findings indicate that WCAG promotes secondary digital wellbeing effects in 3.1% of requirements in WCAG 1.0, rising to 9.3% of requirements in WCAG 2.2 and 17.8% of requirements in WCAG 3.0 (2024-12-12 draft version). Entire subsections of the WCAG 3.0 (draft) requirements were categorized under digital wellbeing, and all of the requirements categorized under digital wellbeing in WCAG 2.2 are captured under W3C's subtheme of cognitive and learning disability. The leading DWB aspects promoted via WCAG were *Avoiding attention-capture dark patterns*, *Supporting meaningful use*, and *Muting distractions to promote productivity*. Future research can further define how existing digital wellbeing methods may be appropriate for users with cognitive and learning disabilities, and how the rigorous automation and standardization approach in WCAG may be applicable to digital wellbeing as well. The presence in WCAG of digital wellbeing aspects which traditionally pertain to the leisure context may also motivate a wider approach of conceiving interventions and frameworks for digital wellbeing in terms of both leisure and non-leisure activities.

2 Background

The design and refinement of consumer digital technology over the past half-century have brought with them immense changes in daily life. These developments, mediated by the widespread use of the personal computer and the smartphone, are characterized by a highly developed set of application design and management strategies which have, at points, disadvantaged users. We present the accessibility and digital wellbeing research areas below, which consider different impacts of the rising use of computers in personal life.

2.1 Accessibility

Accessibility is a property of a system, which can be defined as accessible if it meets certain criteria. The W3C, or World Wide Web Consortium, host the Web Content Accessibility Guidelines (WCAG) and Web Accessibility Initiative (WAI), which define accessibility as: “Accessibility: addresses discriminatory aspects related to equivalent user experience for people with disabilities. Web accessibility means that people with disabilities can equally perceive, understand, navigate, and interact with websites and tools. It also means that they can contribute equally without barriers” [12]. The W3C accessibility definition focuses on the concept of disability, while definitions such as those in ISO Guide 71 (Version 1) are

scoped more widely: Accessible design is “design focused on principles of extending standard design to people with some type of performance limitation to maximize the number of potential customers who can readily use a product, building or service which may be achieved by designing products, services and environments that are readily usable by most users without any modification, by making products or services adaptable to different users (adapting user interfaces), and by having standardized interfaces to be compatible with special products for persons with disabilities” [16].

The WCAG, first released in 1999, are a commonly used series of targeted recommendations which aim to improve the accessibility of web content for people with disabilities [3]. The WCAG are deployed in partial/derivative or full form in over 56 countries as of December 2023, as well as on a supranational level in the European Union, applying in various countries to public and private sector websites [2, 17]. Major social media websites have a mixed track record of compliance with WCAG, with some websites showing very good compliance and some websites demonstrating poor compliance [18].

Over the past 25 years, the number of requirements in each WCAG version has changed from 65 in version 1.0 (1999), to 61 in version 2.0 (2008), to 78 in version 2.1 (2018), to 86 in version 2.2 (2023), to 180 in version 3.0 (draft) [1, 3, 4, 19, 20]. Successive improvements have been made to WCAG, consistently bringing an expanded focus on users with cognitive and learning disabilities [1, 4, 19, 21, 22]. Additionally, the standards have been extended to mobile devices with the 2.1, 2.2, and 2.3 versions [1, 4, 19] and aim to be extended to wearable devices and Internet of Things devices with the 3.0 (draft) version [20]. Nonetheless, each release attempts to preserve former criteria in order to preserve compatibility.

The WAI notes that improved accessibility may cause improved usability as a secondary effect and that there is a “significant overlap between accessibility and usability” [12]. However, usability is only considered relevant in the design of the WCAG to the extent that usability can help disabled users achieve desired goals. Other research has indeed indicated a positive effect on usability based on improved accessibility [13], also noting that accessibility can have negative effects on usability in certain cases [14].

One element of WCAG's accessibility goals for cognitive disabilities is the capacity of the webpage to focus the user's attention on the desired task [5]. This need to support the user's attention span is also noted in WCAG 2.2 literature [23, 24]. Given pressing questions in the wider academic literature concerning how attention is manipulated among users of social media and other leisure websites [8], it is instructive to consider if or how the WCAG specifications have secondary effects on other disciplines of human-computer interaction which focus on attention and time spent, such as the digital wellbeing discipline.

2.2 Digital Wellbeing

Digital wellbeing (DWB) has been conceptualized by Burr et. al in 2020 as “the impact of digital technologies on what it means to live a life that is good for a human being in an information society” [6]. A branch of digital wellbeing research in recent years has focused on behavior change for voluntary, leisure-based use (social media, entertainment, or video apps), framing digital wellbeing in

terms of improving a user’s relationship with their personal use of technology [8, 9]. Digital self-control tools, or DSCTs, are software interventions which users can adopt to improve self-control of their technology use, leveraging strategies such as block/removal, self-tracking, goal-advancement, or reward/punish to modify user interfaces, eventually influencing user time spent, attention, or engagement [7]. These tools have been developed in counterpoint to interests of technology corporations and as an antidote to interfaces that are designed to capture user attention in unwanted ways [8, 15].

As opposed to accessibility or usability, digital wellbeing is considered a supercontextual measurement, extending beyond the immediate context of the actual use of an application or technology, and evaluating mutual influence between different uses, alignment of technology use with the user’s life path, or secondary effects of technology use on third parties or the environment [8]. A recent systematic literature review in the digital wellbeing field has identified aspects, themes, functionality, proposed intervention strategies, behavioral theories and constructs, challenges, relations to ethical principles, evaluations, participant background in studies, evaluation measures and scales, and general effectiveness [8]. Of particular interest for this study are the six digital wellbeing aspects identified in [8] which characterize key user-facing effects.

One of these six digital wellbeing aspects refers to avoiding *attention-capture damaging patterns* (also sometimes called *attention-capture dark patterns* or ACDPs). Based on the concept of the “dark pattern”, Monge Roffarello 2023 defines ACDPs as “recurring patterns that designers adopt to manipulate users into spending attention in ways that often lead to a loss of sense of control and time, and feelings of regret” [15]. Further research has indicated the relevance of surfacing these ACDPs to users as they use a technology [25].

Future research directions in digital wellbeing include proactive support for digital wellbeing, more smooth integration of DSCTs into usage patterns, better theoretical measures, studies of DSCTs in the long-term context, stronger ethical models, and improved stakeholder involvement [8].

3 Data Collection and Analysis

This study aims to understand to what extent the WCAG specifications have a secondary effect on promoting digital wellbeing and avoiding attention-capture dark patterns, as defined in [8] and [15]. Three researchers in digital wellbeing design of consumer applications participated in the study, and each classified each of the requirements in the three major versions of the WCAG with respect to secondary effects on digital wellbeing. Two of the researchers had additional expertise in design of government technology, and two had additional expertise in conducting accessibility evaluations. The classification was conducted among the six digital wellbeing aspects from [8], rephrased in a positive verb form: *Reducing digital overuse*, *Muting distractions to promote productivity*, *Supporting meaningful use*, *Avoiding attention-capture dark patterns*, *Creating a positive environmental footprint*, *Avoiding dangerous habits*, as well as *Other* (leaving open the possibilities for “None” as an option and for each requirement to be classified in multiple categories). A space for additional notes was also provided for each researcher to justify their response: if *Avoiding attention-capture dark patterns* was selected, the reviewer provided which ACDP(s) of the 11 options in [26] were affected. Each of the basic requirements of the latest major versions of WCAG (1.0, 2.2 and 3.0 draft version) were evaluated (all “Checkpoints” in WCAG 1.0, all “Success Criteria” in WCAG 2.2, and all “Requirements/Assertions” in WCAG 3.0 (draft) [3, 4, 20]). After separately coding all requirements, researchers met to confirm a final coding for each requirement. Given the exploratory nature of the coding, researchers were encouraged to resolve disagreements in the final discussions by adopting a narrow interpretation of DWB and ACDP classifications. Final coding data is provided as a supplementary paper attachment (CSV file).

4 Quantitative Results

The quantitative results from the expert coding are presented in this section, considering the overall classification (Figure 1), the six secondary digital wellbeing aspects stemming from the WCAG accessibility requirements (Table 1), as well as the eleven ACDPs

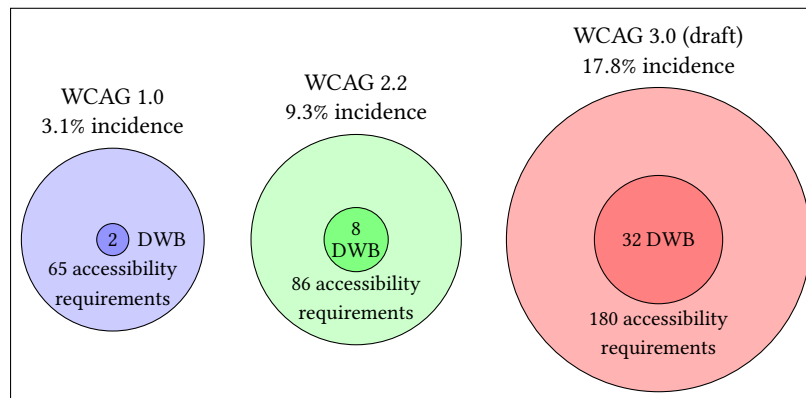


Figure 1: Number and proportion of accessibility requirements promoting secondary digital wellbeing (DWB) effects in WCAG 1.0, WCAG 2.2, and WCAG 3.0 (draft) specifications. The accessibility requirements promoting digital wellbeing increase in number and proportion over time.

Table 1: Distribution of secondary DWB aspects ([8]) in WCAG 1.0, WCAG 2.2, and WCAG 3.0 (draft) specifications.

	Avoiding attention-capture dark patterns	Supporting meaningful use	Muting distractions to promote productivity	Reducing digital overuse	Creating a positive environmental footprint	Avoiding dangerous habits	Total
WCAG 1.0	1	2	0	0	0	0	3
WCAG 2.2	6	0	3	0	1	0	10
WCAG 3.0 (2024-12-12 draft)	23	10	4	2	0	1	40
Total	30	12	7	2	1	1	53

which were relevant for the specific digital wellbeing aspect *Avoiding attention-capture dark patterns* (Table 2).

The three major WCAG versions, when considered as a group, were found to have a secondary effect on all of the six DWB aspects from the 2023 digital wellbeing systematic review [8] (Table 1). Multiple secondary DWB effects were found for some accessibility requirements (ex. the case of 2.2.2 *Pause, Stop, Hide* in WCAG 2.2, which had a secondary impact on *Muting distractions to promote productivity* and *Avoiding attention-capture dark patterns*). 2 requirements from WCAG 1.0 (3.1%) were noted to promote 3 secondary DWB effects, 8 requirements from WCAG 2.2 (9.3%) were noted to promote 10 secondary DWB effects, and 32 requirements from WCAG 3.0 draft version (17.8%) were noted to promote 40 secondary DWB effects (see Figure 1 and Table 1). Due to the relatively open coding method used and high proportion of negative ratings, Gwet’s AC1 [27] was used to measure inter-rater reliability on the binary presence/absence of secondary DWB effects, yielding a score of 0.698 (“Substantial” agreement strength according to the kappa-type interpretations from [28]).

Out of the 11 ACDPs from [15] composing the digital wellbeing aspect of *Avoiding attention-capture dark patterns* (Table 2), the most common ACDPs avoided as a secondary effect of the WCAG were

Attentional Roach Motel (13 instances in total among all WCAG versions analyzed), *Disguised Ads and Recommendations* (10 instances in total among all WCAG versions analyzed) and *Neverending Autoplay* (5 instances in total among all WCAG versions analyzed). Notably, *Infinite Scroll* and *Time Fog* were not identified as ACDPs avoided as a secondary effect of the WCAG specifications (Table 2).

Of note is the increase over time (from WCAG 1.0 to 2.2 to 3.0 draft version, Figure 1) in the number and proportion of accessibility requirements which have secondary effects on digital wellbeing. This is in part due to increased control of timing and interruptions in the WCAG 2.0-2.2 standards, and due to novel accessibility sections in the 3.0 (draft) version creating a new focus on misinformation, deception, and algorithmic control. Within WCAG 3.0 (draft), the entire sections 2.9.2 *Adequate Time* (3 requirements), 2.9.3 *Unnecessary Steps* (2 requirements), 2.9.4 *Avoid Deception* (5 requirements) and 2.12.5 *Control Interruptions* (1 requirement) were classified as having a secondary effect on digital wellbeing.

The W3C notes seven subcategories in WCAG 2.2 as particularly relevant to accessibility for cognitive disabilities [5]: all of the digital wellbeing secondary effects in WCAG 2.2 were classified under four of these subcategories (1.4 *Distinguishable*, 2.2 *Enough Time*, 3.1 *Readable*, 3.3 *Input Assistance*) while the remaining three subcategories mentioned by W3C (1.3 *Adaptable*, 2.4 *Navigable*,

Table 2: Distribution of attention-capture dark patterns ([15]) for the *Avoiding attention-capture dark patterns* digital wellbeing aspect in WCAG 1.0, WCAG 2.2, and WCAG 3.0 (draft) specifications.

	Attentional Roach Motel	Disguised Ads and Recommendations	Neverending Autoplay	Fake Social Notifications	Guilty Pleasure Recommendations	Playing by Appointment	Recapture Notifications	Casino Pull-to-refresh	Grinding	Infinite Scroll	Time Fog	Total
WCAG 1.0	1	0	0	0	0	0	0	0	0	0	0	1
WCAG 2.2	2	0	2	0	0	1	1	0	0	0	0	6
WCAG 3.0 (2024-12-12 draft)	10	10	3	4	4	3	2	1	1	0	0	38
Total	13	10	5	4	4	4	3	1	1	0	0	45

3.2 *Predictable*) do not contain any secondary digital wellbeing aspects. No digital wellbeing requirements in WCAG 2.2 were present outside of these subcategories for cognitive disability.

5 Qualitative Results

After completing the quantitative study, the researchers recorded a list of additional trends captured throughout the expert coding process (both during the final coding discussion and based on individual notes on each WCAG requirement). The following subsections discuss these trends: (1) positive secondary effects on digital wellbeing, which were the main focus of the quantitative study; (2) enabling secondary effects on digital wellbeing, which do not promote digital wellbeing directly but provide the technical capacity to support digital wellbeing; (3) limited secondary effects on digital wellbeing, which refer to WCAG requirements that could potentially support digital wellbeing but as worded do not do so; and (4) negative secondary effects on digital wellbeing.

5.1 Positive Secondary Effects of WCAG on Digital Wellbeing

The quantitative study results were found to be supportive of digital wellbeing. Within the 2 of 65 requirements (WCAG 1.0), 8 of 86 requirements (WCAG 2.2) and 32 of 180 requirements (WCAG 3.0 draft) which were found to promote positive secondary digital wellbeing effects (Table 1), two requirements were noted as being unusually relevant to digital wellbeing: the WCAG 3.0 (draft) requirements 2.9.3 *Optional Information* and 2.9.3 *Optional Input* may have particularly transformational effects on digital wellbeing if unnecessary content or information is completely eliminated from web interfaces. Although compliance with both requirements would help avoid 6 out of the 11 ACDPs (*Guilty Pleasure Recommendations*, *Disguised Ads and Recommendations*, *Recapture Notifications*, *Grinding*, *Attentional Roach Motel*, and *Fake Social Notifications*), compliance would likely have no effect on the other ACDPs (*Infinite Scroll*, *Casino Pull-to-refresh*, *Neverending Autoplay*, *Playing by Appointment*, and *Time Fog*).

5.2 Enabling Secondary Effects of WCAG on Digital Wellbeing

Successful DSCT software development (i.e. the programmatic recognition and modification of page elements to promote digital wellbeing) is enabled by basic webpage structure standardization in the WCAG specification (such as WCAG 2.2 1.3.1 *Info and Relationships*) even if these accessibility requirements do not explicitly cause digital wellbeing.

In WCAG 2.2, 1.3.6 *Identify Purpose* goes beyond this level of underlying structure to enable DSCTs to potentially recognize and remove unnecessary elements. In WCAG 3.0 (draft), 2.3.4 *Control Location*, 2.7.1 *Distinguishable Sections*, and 2.12.3 *Content Markup* can also help indicate which elements are most relevant to users, allowing DSCTs to remove unneeded elements. 2.7.2 *Familiar Layout* and 2.11.1 *New Interfaces* may also help reduce the software development overhead and maintenance costs of DSCTs, enabling interventions to react to similar ACDPs in similar interfaces. 2.9.1 *Allow Automated Entry* would also enable DSCTs to complete tasks

on behalf of all users, building on a longstanding software development trend of leveraging accessibility features for a wider group of users [29–32].

5.3 Limited Secondary Effects of WCAG on Digital Wellbeing

The accessibility requirements contained in the WCAG specifications can be considered in some cases to have limited effects on digital wellbeing because they don't imply a necessary condition that would be supportive of digital wellbeing.

In WCAG 3.0 (draft), requirement 2.12.1 *Text Customization* uses the word “can”, failing to concretely establish how easy it is to customize text, or the way in which the text can be changed, meaning that the actual user-facing improvement may not be created. 3.3.7 *Redundant Entry* in WCAG 2.2 and 2.9.5 *Redundant Entry* in WCAG 3.0 (draft) might help avoid the *Attentional Roach Motel* ACDP if they were to extend between processes (see Item 5 in [33]), but as-is (limited to single processes) are not restrictive enough. In WCAG 3.0 (draft), 2.4.2 *Inform Before Activation*, 2.12.1 *Adjust Background* and 2.12.6 *Visual Stimulation* are also not specific enough to have a clear impact on DWB aspects such as *Reducing digital overuse* and the *Infinite Scroll* ACDP.

Additionally, the A/AA/AAA distinction between levels of conformance in WCAG 2.X versions may not be correlated directly to digital wellbeing: notable is 3.3.4 *Error Prevention (Legal, Financial, Data)* (Level AA) in WCAG 2.2 which focuses on database contents and could help prevent auto-subscriptions or accidental subscriptions as entailed in the *Attentional Roach Motel* ACDP. However, the more widely scoped 3.3.6 *Error Prevention (All)* (Level AAA), which refers to a different modality (information delivered in user-initiated form submissions), was not classified as having a secondary DWB effect.

5.4 Negative Secondary Effects of WCAG on Digital Wellbeing

The researchers noted that the accessibility guidelines contained in the WCAG may worsen digital wellbeing to some extent as a general outcome: by following best design practices from WCAG, otherwise badly designed leisure interfaces (e.g. social media or video sites) may in fact become more attention-capturing because they become more usable (see [34]); secondly, accessibility features may also expand the user base of such leisure sites to a wider population of individuals.

In terms of specific negative effects on digital wellbeing, 2.9.2 *Adequate Time* in WCAG 3.0 (draft) can limit the ability of DSCTs to set timeouts on use, potentially weakening the effectiveness of digital wellbeing technology unless an exception is made for essential use (as in 2.2.1 *Timing Adjustable* in WCAG 2.2). In WCAG 3.0 (draft) 2.3.4 *Conventions* may also limit DSCT effectiveness due to the unconventional interventions that DSCTs sometimes use to block or limit interface functionality [7]. In WCAG 3.0 (draft), 2.3.4 *Consistent Visual Design* may encourage unwanted design elements (such as buttons, advertising or misleading posts) to be designed similarly to desired content, increasing user manipulation.

6 Limitations

This paper considers WCAG as a way to understand broader implications and connections between accessibility and digital wellbeing, given the wide implementation and development of the WCAG standards. Although the WCAG 1.0 and 2.2 versions are considered official, WCAG 3.0 is in draft form as of March 2025 [20], so its implementation and structure cannot be considered final. Additionally, the WCAG standards are not implemented across all websites and do not apply to non-web applications, which make up a common usage context for mobile devices. Further, only ACDPs and DWB aspects typically arising from the *leisure* digital wellbeing context were considered in the accessibility requirements expert coding, due to the nature of current research in the digital wellbeing field.

7 Future Work

Given the increasing overlap found between WCAG and digital wellbeing aspects, one potential future area of study could be the extent to which individuals with cognitive or learning disabilities would benefit from the digital wellbeing approach and/or DSCTs. Successfully building on the existing research domain of web extensions and tools for cognitive and learning disabilities [35–38], studies could be undertaken on the wide range of existing DSCT apps [15] to evaluate their suitability for helping users with cognitive and learning accessibility needs complete their goals. Noting this possibility, not all ACDPs were found in the WCAG space (*Infinite Scroll* and *Time Fog* were not found). Further research may be considered in the draft of WCAG 3.0 to identify if these ACDPs also pose risks to individuals with cognitive or learning disabilities, and if so, how design to prevent ACDPs might be integrated into the final WCAG 3.0 specification.

Noting the increasing applicability of digital wellbeing concepts to WCAG accessibility requirements, it may be relevant to pursue a wider vision of digital wellbeing and digital self-control which does not focus on leisure behavior (“overuse”) alone, but also considers how digital wellbeing may be promoted in non-leisure behavior.

Another future area of focus could be evaluating the extent to which the WCAG can be used as a model or conceptual reference for standardizing digital wellbeing functionality. Context-specific programmatic evaluation of user flows and interfaces has been developed in existing dark patterns literature [39] but has not been flexible enough to fully extend fully automatic interface recognition and intervention to the DSCT space, although there is a need noted in current DSCT literature for proactive DSCT intervention [8, 40]. Also of interest is the potential for digital wellbeing standardization for websites themselves (i.e. without requiring user-initiated interventions). Building on the existing research theme of automated accessibility checkers [41, 42], WCAG requirements (or other requirements) relevant to DWB could be checked on an automated basis, allowing websites to be “rated” according to their level of digital wellbeing.

8 Conclusion

Accessibility guidelines and digital wellbeing aspects are subjects of increasing recent research interest, as shown by the development of WCAG 3.0 (draft) and a recent systematic review of digital wellbeing [8, 20]. This research work indicates their increasing overlap, often

characterized by attention capture in digital wellbeing research and attentional support for cognitive and learning disabilities in accessibility research. Both the accessibility and digital wellbeing research areas within human-computer interaction focus on expanding the possibilities of the human lifespan, and may lead to productive mutual effects on each other, further promoting positive human interaction with technology.

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